



Shape from polarization via a physical prior-based deep fusion network with ambiguous surface normals

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Abstract: Shape from polarization imaging is a passive three-dimensional imaging method with high precision and detailed reconstruction capabilities, widely used in fields such as intelligent manufacturing, surface defect detection, and medical imaging. However, existing deep learning-based shape from polarization methods have not effectively fused polarization and other physical prior information, leading to degraded reconstruction quality in complex scenes. To address this issue, we propose an innovative method that integrates polarization information with ambiguous surface normals and specular confidence information. By incorporating these additional prior features, the model's reconstruction accuracy in complex scenes is significantly improved. This research calculates the ambiguous surface normals and specular confidence information based on a physical model and introduces a novel dual-branch deep fusion network. The network efficiently extracts multi-scale feature information through a feature extraction module and effectively fuses polarization information with ambiguous surface normals and specular confidence information via a feature fusion module, enhancing the reconstruction accuracy in complex scenes. Experimental results demonstrate that the proposed method can accurately reconstruct surface normal under complex lighting and low-texture conditions, significantly improving the accuracy and robustness of shape from polarization methods. The method is expected to have broad applications in intelligent manufacturing, surface defect detection, and medical imaging. Our dataset and source code will be publicly available at <https://github.com/singobl/CGA-Transformer>.

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1. Introduction

Shape from polarization (SfP), as an important passive three-dimensional (3D) imaging technique, has garnered significant attention in recent years due to its widespread applications in industrial component dimensional measurement, surface defect detection, and medical imaging [1]. By analyzing the interaction between polarized light and object surfaces, SfP estimates surface normal and geometric information [2], making it particularly suitable for 3D reconstruction tasks in complex scenarios involving low texture [3], strong specular reflections, and transparent objects [4]. Compared with active imaging technologies such as LiDAR [5] and structured light [6], SfP is not constrained by imaging distance and avoids issues like sparse data generation and noise sensitivity, thus exhibiting significant potential in complex environments [7].

Early SfP methods primarily relied on Fresnel equations and polarized reflection physics models [8,9], inferring 3D surface structures by measuring the azimuth and zenith angles of polarized light. Notable implementations include Rahmann et al. [10] approach under

pure specular reflection conditions and Miyazaki et al. [11] technique utilizing a pure diffuse reflection model for surface normal estimation. However, these methods necessitated stringent assumptions regarding material properties and reflection types (specular/diffuse), leading to practical limitations including π -radians azimuthal ambiguity (manifested as convex-concave inversion artifacts) and zenith angle estimation inaccuracies. To address the above issues, researchers have explored methods that integrate additional information sources with SfP, including SfP combined with depth data (e.g., through binocular stereo vision [12] or time-of-flight (ToF) camera [13] sensors for coarse depth estimation) and SfP combined with illumination [14] or shading cues [15]. Nevertheless, such hybrid approaches remain constrained by challenges in multi-sensor registration precision and operational dependencies on specialized imaging configurations requiring controlled lighting/shadow projection environments. These inherent limitations significantly impede their applicability in dynamic real-world scenarios demanding flexible acquisition setups.

In recent years, deep learning-based SfP methods have significantly improved the robustness of 3D reconstruction. Ba et al. [16] incorporated multiple physically modeled ambiguous surface normals as prior inputs into a U-Net network, guiding the network to derive optimal solutions from polarization information. Kondo et al. [17] synthesized data based on a polarized BRDF model and employed a convolutional neural network to enhance the robustness of normal reconstruction. Lei et al. [3] proposed a multi-view encoding approach combined with a Transformer architecture to handle complex scenes, while Shao et al. [4] developed a specialized SfP method for transparent objects. The core breakthrough in current research lies in the collaborative modeling of multiple physical priors. For instance, Liu et al. [7] were the first to integrate shading cue and specular confidence features into a network, achieving improved normal estimation in multi-light scenarios through deep fusion. However, these methods still face critical limitations: they rely on post-processing optimization rather than raw polarization signals, failing to fundamentally resolve the ambiguity in normal estimation. The root cause is that existing approaches have not achieved deep integration of polarization texture features with physical model constraints (e.g., reflection type discrimination, ambiguous surface normals constraints) through deep learning mechanisms, preventing the network from autonomously suppressing ill-posed solutions.

To address the above challenges, we propose a novel method that fuses polarization information with ill-posed physical models, incorporating ambiguous surface normals and a specular confidence mechanism specifically designed for highly reflective regions. To efficiently fuse these features in complex scenes, we design a novel dual-branch deep fusion network, which introduces a coarse-to-fine hybrid fusion scheme based on content-guided attention (CGA) [18] and a Transformer Encoder module [19] for global feature extraction. Extensive experiments conducted under challenging conditions-including multi-illumination, low-texture, and strong specular reflections-demonstrate that our method outperforms existing SfP approaches in both qualitative and quantitative evaluations.

2. Proposed method

In this chapter, the polarization and ambiguous surface normal are elaborated in detail, along with the specular confidence information, with particular emphasis on the significance and computation of ambiguous surface normals as well as specular confidence.

2.1. Polarization priors

When the surface of a dielectric material is illuminated by unpolarized light, the reflected light becomes partially polarized [20]. By placing a linear polarizer with a polarization angle ϕ_{pol} in front of the camera (as shown in Fig. 1(b)), the intensity of the captured image pixels under

unpolarized illumination follows a sinusoidal variation [21], as expressed by Eq. (1):

$$I(\phi_{pol}) = I_{un}[1 + \rho \cos(2(\phi_{pol} - \phi))], \quad (1)$$

Here, ϕ represents the angle of polarization (AoP), ρ denotes the degree of polarization (DoP), and I_{un} signifies the intensity of unpolarized light. We define $I_{0^\circ}, I_{45^\circ}, I_{90^\circ}, I_{135^\circ}$ as the light intensities captured by the camera through polarizers oriented at angles of $0^\circ, 45^\circ, 90^\circ, 135^\circ$, respectively. Based on the Fresnel equations, the calculations for the AoP ϕ , the DoP ρ , and the intensity of unpolarized light I_{un} are presented in Eqs. (2)–(4), respectively:

$$\phi = \frac{1}{2} \arctan \left(\frac{I_{45^\circ} - I_{135^\circ}}{I_{0^\circ} - I_{90^\circ}} \right), \quad (2)$$

$$\rho = \frac{\sqrt{(I_{0^\circ} - I_{90^\circ})^2 + (I_{45^\circ} - I_{135^\circ})^2}}{I_{0^\circ} + I_{90^\circ}}, \quad (3)$$

$$I_{un} = (I_{0^\circ} + I_{45^\circ} + I_{90^\circ} + I_{135^\circ})/2, \quad (4)$$

As illustrated in Fig. 2(b), I_{un} is employed to represent the overall two-dimensional texture information of the object's surface, while the DoP ρ (Fig. 2(c)) and AoP ϕ (Fig. 2(d) colored in the HSV space) are associated with the two angular components of the surface normal, thereby encapsulating the surface's geometric information. Additionally, by analyzing the term $\cos(2(\phi_{pol} - \phi))$ in Eq. (1), a π -radian ambiguity in the phase angle ϕ is identified: when two phase angles differ numerically by π , the resulting polarization image intensity variations become identical, leading to the pseudo-concavity issue as depicted in Fig. 1(a), where N and N' exhibit significant directional differences corresponding to convex and concave surfaces, respectively. To overcome this problem, Lei et al. [3] proposed a polarization encoding strategy that is more efficient than existing phase angle representations by encoding the original phase angle ϕ to eliminate its drawbacks. Therefore, the phase angle is encoded using $\phi_e = (\cos(2\phi), \sin(2\phi))$, as illustrated in Fig. 2(e), and the surface geometric characteristics are more accurately described by the re-encoded phase angle ϕ_e . Furthermore, we introduce the viewing encoding V [3] (as illustrated in Fig. 2(i)) to describe the observation direction corresponding to each pixel during the imaging process. Based on the above analysis, the polarization prior information in this study can be defined as $Pol_{prior} = [images; I_{un}; \rho; \phi_e; V]$. The variable V is provided by the dataset; if the dataset does not include this information, we use normalized 2D coordinates as a substitute.

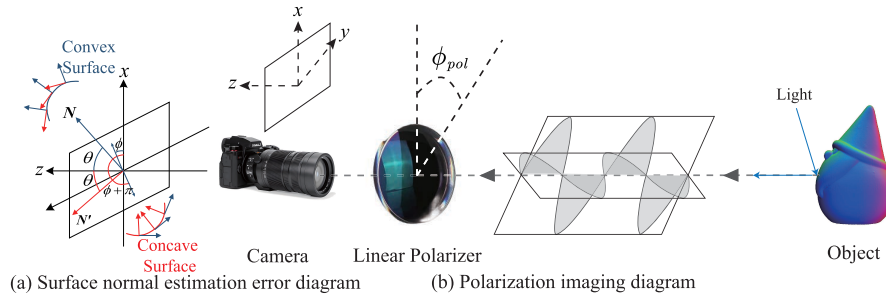


Fig. 1. Surface normal and polarization imaging diagram.

2.2. Ambiguous surface normals

In SfP techniques, relying solely on polarization priors often proves insufficient to effectively resolve the π -ambiguity in the azimuth angle and the uncertainty in zenith angle estimation under

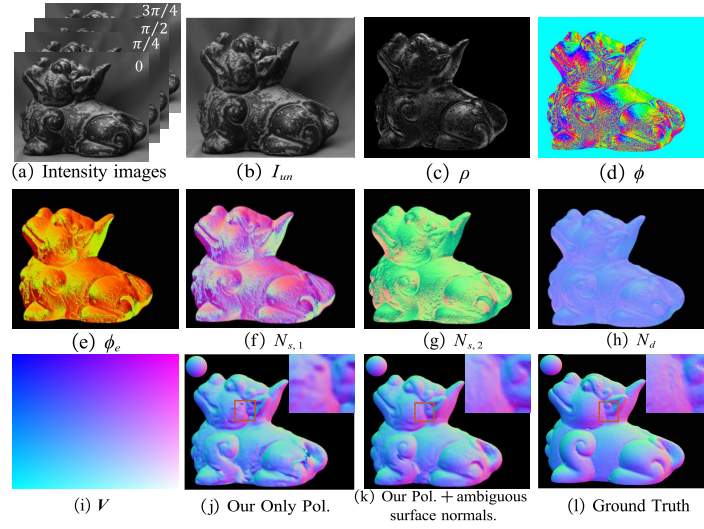


Fig. 2. Importance of ambiguous surface normals on DeepSfP dataset [16]. The spheres inserted into the ground truth are color-visualized to show the direction of the surface normal.

complex scenarios. To address this, the present study further incorporates ambiguous surface normal derived from ill-posed physical models, thereby enhancing the network's robustness and accuracy in challenging conditions such as multi-light sources and multiple material properties.

In a typical SfP task, the surface normal $N = [\cos \varphi \sin \theta; \sin \varphi \sin \theta; \cos \theta]$ can be described by the azimuth angle φ and the zenith angle θ [22]. According to the Fresnel equations, when the diffuse reflection component predominates, the DoP ρ_d can be expressed in terms of the refractive index n and the zenith angle θ_d , as shown in Eq. (5) [22]:

$$\rho_d = \frac{(n - \frac{1}{n})^2 \sin^2 \theta_d}{2 + 2n^2 - (n + \frac{1}{n})^2 \sin^2 \theta_d + 4 \cos \theta_d \sqrt{n^2 - \sin^2 \theta_d}}, \quad (5)$$

Moreover, because ρ_d has a relatively weak dependence on n [22], we assume $n = 1.5$ throughout the remainder of this paper. With this known n , Eq. (5) can be rearranged to yield an approximate form of the zenith angle under diffuse-dominant conditions, and a numerical solver is then employed to obtain the approximate zenith angle estimate. Under such diffuse-dominant conditions, the relationship between the azimuth angle φ_d and AoP ϕ is given by $\varphi_d = \phi$ [14]. We obtain the ambiguous surface normals under dominant diffuse reflection conditions by calculating the zenith angle θ_d through Eq. (5) and the azimuth angle φ_d , as formally expressed $N_d = [\cos \varphi_d \sin \theta_d; \sin \varphi_d \sin \theta_d; \cos \theta_d]$.

$$\rho_s = \frac{2 \sin^2 \theta_s \cos \theta_s \sqrt{n^2 - \sin^2 \theta_s}}{n^2 - \sin^2 \theta_s - n^2 \sin^2 \theta_s + 2 \sin^4 \theta_s}, \quad (6)$$

When specular reflection is dominant, the relationship in Eq. (6) must be employed, and the zenith angle θ_s is obtained via interpolation [22]. However, in the absence of external constraints, solving Eq. (6) can yield two possible solutions $(\theta_{s,1}, \theta_{s,2})$. The relationship between the azimuth angle φ_s and AoP ϕ is given by $\varphi_s = \phi + \frac{\pi}{2}$ [14]. Finally, we derive two ambiguous surface normal vectors under dominant specular reflection, with their expressions given as $N_{s,1} = [\cos \varphi_s \sin \theta_{s,1}; \sin \varphi_s \sin \theta_{s,1}; \cos \theta_{s,1}]$ and $N_{s,2} = [\cos \varphi_s \sin \theta_{s,2}; \sin \varphi_s \sin \theta_{s,2}; \cos \theta_{s,2}]$.

The importance of ambiguous surface normals. Unlike polarization priors, ambiguous surface normals can directly provide disambiguation cues for the uncertainties caused by

azimuthal π -ambiguity and zenith angle variations. Therefore, ambiguous surface normals contain 3D physical information complementary to polarization priors. Fig. 2(f) and Fig. 2(g) show the ambiguous surface normals computed based on the specular reflection model, while Fig. 2(h) presents those derived from the diffuse reflection model. These three sets of ambiguous surface normals contain rich geometric features, providing crucial information for surface normal reconstruction. Fig. 2 demonstrates an example of the benefits of fusing polarization priors with ambiguous surface normals. When only polarization priors are considered, the reconstructed surface normals exhibit significant errors (as shown in Fig. 2(j)). However, when ambiguous surface normals are incorporated, the reconstruction is notably improved (see Fig. 2(k)).

2.3. Specular confidence

In complex scene reconstruction tasks, large specular-reflection areas frequently appear—for example, the wall in Fig. 3 contains multiple extensive specular-reflection regions, making it difficult to accurately estimate surface normal in these areas. Liu et al. [7] introduced specular confidence prior information, which significantly improved reconstruction accuracy for regions with strong specular reflections. As illustrated in Fig. 3, an importance experiment was carried out to evaluate the specular confidence feature. Although the DoP ρ in the polarization prior (Fig. 3(c)) also appears larger in regions with strong specular reflections, the object's material and other factors affect the accuracy of specular separation. From Fig. 3(b) and Fig. 3(c), it can be observed that DoP alone cannot accurately delineate specular reflection areas, and many specular regions on the wall can only be correctly identified using the specular confidence H . From the visual comparisons in Fig. 3, it is evident that the surface normal estimated without specular confidence information (Fig. 3(d)) exhibit marked reconstruction errors in the numerous strongly specular patches on the wall, resulting in lower reconstruction quality compared to the results obtained when specular confidence is included (Fig. 3(e)). Following the method presented in [7], we compute the specular confidence H based on the intensity variations observed in four original polarization images, as shown in Eq. (7):

$$H = \min_{y \in I_{un}} \{ \max \{ I_{0^\circ}, I_{45^\circ}, I_{90^\circ}, I_{135^\circ} \} - \min \{ I_{0^\circ}, I_{45^\circ}, I_{90^\circ}, I_{135^\circ} \} \}, \quad (7)$$

In our experiments, the pixel-block size y is set to 3×3 by default.

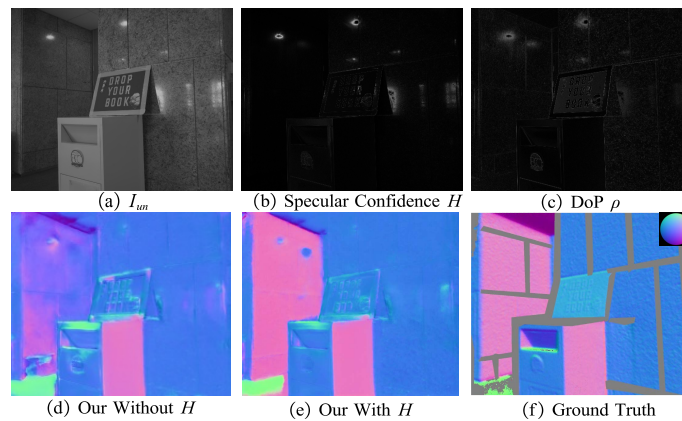


Fig. 3. Importance of the Specular Confidence H Experiment. The spheres inserted into the ground truth are color-visualized to show the direction of the surface normal.

3. Dual-branch deep fusion network

Building on polarization prior, ambiguous surface normals, and specular confidence information, this chapter introduces a CGA-Transformer two-branch network architecture designed to estimate surface normal. As shown in Fig. 4. First, we introduce the overall network architecture in detail. Next, we describe the CGA-based hybrid fusion strategy [18]. Finally, we present the multi-head attention mechanism [19], followed by the loss function that constrains network training.

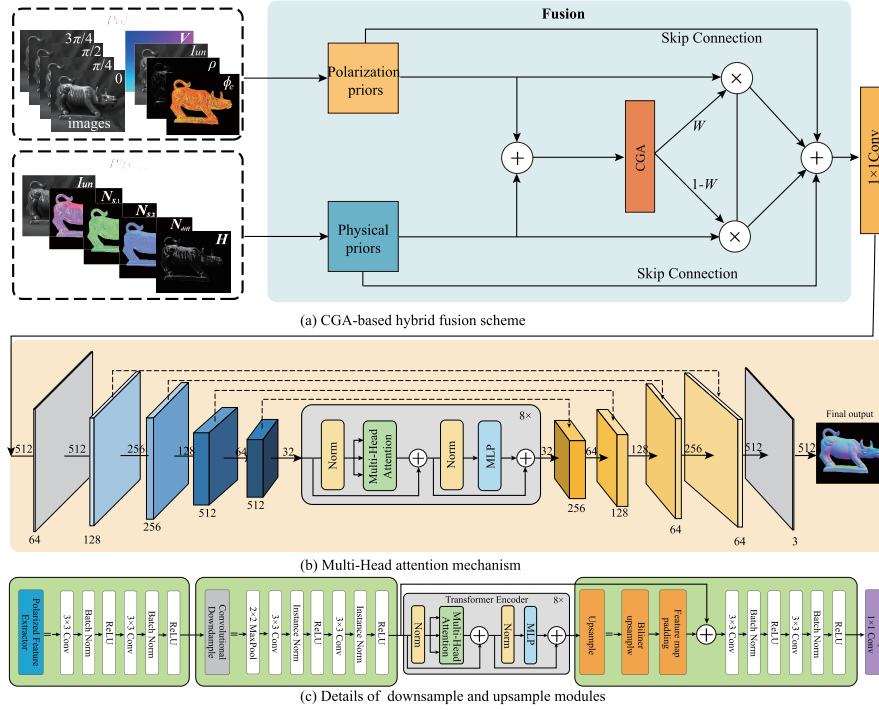


Fig. 4. Overall dual-branch deep fusion network architecture.

3.1. Overall architecture

Our goal is to estimate the surface normals based on a fusion of polarization and ambiguous surface normals and specular confidence information. To this end, we propose a dual-branch deep fusion network, as shown in Fig. 4. The polarization prior $Pol_{prior} \in \mathbb{R}^{H \times W \times 11}$ consists of five components: polarization images $\in \mathbb{R}^{H \times W \times 4}$, unpolarized intensity $I_{un} \in \mathbb{R}^{H \times W \times 1}$, phase angle $\phi_e \in \mathbb{R}^{H \times W \times 2}$, and viewpoint encoding $V \in \mathbb{R}^{H \times W \times 3}$. The physical model prior $Phy_{prior} \in \mathbb{R}^{H \times W \times 11}$ comprises four parts: two surface normals vectors $N_{s,1} \in \mathbb{R}^{H \times W \times 3}$ and $N_{s,2} \in \mathbb{R}^{H \times W \times 3}$ calculated based on the specular reflection model, one surface normal vector $N_d \in \mathbb{R}^{H \times W \times 3}$ calculated based on the diffuse reflection model, unpolarized intensity $I_{un} \in \mathbb{R}^{H \times W \times 1}$, and specular confidence information $H \in \mathbb{R}^{H \times W \times 1}$. As illustrated in Fig. 4, the CGA-Transformer consists of a CGA-based hybrid fusion scheme, an encoder, a Transformer Encoder, and a decoder. The CGA-based hybrid fusion scheme aims to extract and fuse the polarization prior with ambiguous surface normals and specular confidence prior features. The multi-head attention mechanism is utilized to capture global contextual information, thereby enhancing the expressive capability of the features. Finally, high-quality surface normal maps are obtained through the upsampling operation of the decoder.

3.2. CGA-based hybrid fusion scheme

In this study's SFP task, our goal is to fuse polarization information with ambiguous surface normals and specular confidence features to accurately reconstruct surface normal. To improve the efficiency of feature extraction, we introduce a CGA-based hybrid fusion scheme. The core idea is to adaptively modulate the features through the learned spatial and channel weights. Figure 4(a) shows the details of a CGA-based hybrid fusion scheme. The core part is to use CGA to enhance feature representation capabilities.

Content-guided attention mechanism. Conventional feature attention modules [23] are restricted by the lack of interaction between channel and spatial attention, as well as their inability to capture anything beyond image-level information distribution, resulting in constrained feature extraction capabilities in complex scenes. To address this, we adopt a coarse-to-fine CGA mechanism that generates unique spatial importance maps (SIM), i.e., $W \in \mathbb{R}^{C \times H \times W}$ for each channel, enabling dynamic interaction between channel and spatial dimensions. Figure 5(a) illustrates the implementation details of CGA. As formulated in Eqs. (8)–(9), CGA generates channel attention weights W_c and spatial attention weights W_s .

$$W_c = C_{1 \times 1}(\max(0, C_{1 \times 1}(X_{GAP}^c))), \quad (8)$$

$$W_s = C_{7 \times 7}([X_{GAP}^s, X_{GMP}^s]), \quad (9)$$

where $\max(0, x)$ denotes the ReLU activation function, $C_{k \times k}(\cdot)$ represents a convolutional operator with a $k \times k$ kernel, and $[\cdot]$ indicates a channel-concatenation operation. Let X_{GAP}^c , X_{GAP}^s , and X_{GMP}^s represent the features obtained via global average pooling across the channel dimension, global average pooling across the spatial dimension, and global max pooling across the spatial dimension, respectively. Subsequently, we fuse W_c and W_s via a simple element-wise addition following broadcast rules to obtain a coarse SIM $W_{coa} \in \mathbb{R}^{C \times H \times W}$, as shown in Eq. (10):

$$W_{coa} = W_c + W_s, \quad (10)$$

where W_{coa} and X are aligned by channel, since W_c is channel-based. To obtain the final refined SIM W , each channel of W_{coa} is further adjusted based on the corresponding input features. We leverage the content of the input features to guide the generation of the final channel-specific SIM W . Specifically, as shown in Eq. (11), the channel shuffle ($CS(\cdot)$) operation is employed to alternately rearrange the channels of W_{coa} and X . When combined with convolutional layers, this approach significantly reduces the number of parameters while maintaining expressive power.

$$W = \sigma(\mathcal{GC}_{7 \times 7}(CS([X, W_{coa}]))), \quad (11)$$

Here, σ denotes the sigmoid operation, $CS(\cdot)$ represents the channel shuffle operation, and $\mathcal{GC}_{k \times k}(\cdot)$ denotes a group convolution layer with a kernel size of $k \times k$. Figure 5(b) displays four representative feature maps (colored in the HSV space) extracted from the 11-channel SIM W . It can be observed that the two feature maps in the first row contain important feature information of high-reflective regions and geometric textures, while the two feature maps in the second row contain rich geometric features of surface normal and viewpoint-encoding feature. The final feature (i.e., F_{fuse}) computation process is shown in Eq. (12).

$$F_{fuse} = C_{1 \times 1}(Phy_{prior} \cdot W + Pol_{prior} \cdot (1 - W) + Phy_{prior} + Pol_{prior}), \quad (12)$$

where $\mathbf{1}$ represents a matrix with all one elements and is used to implement the corresponding addition operation, and $C_{1 \times 1}(\cdot)$ represents a 1×1 convolution layer.

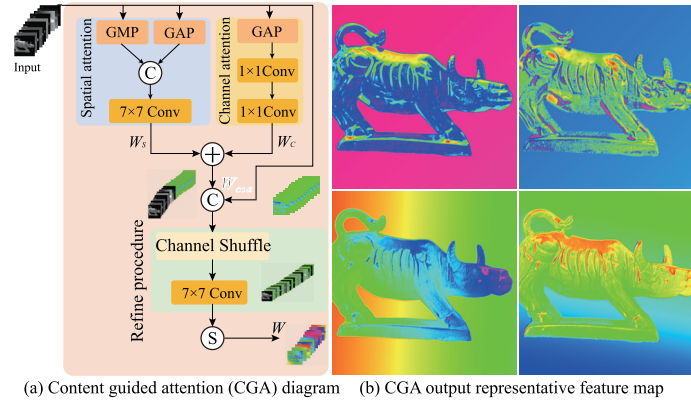


Fig. 5. Content-guided attention mechanism diagram.

3.3. Multi-head attention mechanism

To further process the features fused by CGA, we introduce a multi-head attention mechanism [19] in the intermediate part of the encoder-decoder structure to utilize global contextual information (as shown in Fig. 4(b)). Unlike similar methods that combine a CNN [24] encoder with a Transformer [25], since CGA has already embedded and integrated the features, we omit the linear projection layers typically found in traditional Transformers. Additionally, viewing encoding V provides information similar to positional embeddings, so we do not introduce positional embeddings separately.

3.4. Loss function

During the training phase, we employ cosine similarity loss as our loss function [16], as presented in Eq. (13).

$$L_{cos} = \frac{1}{W \times H} \sum_i^W \sum_j^H (1 - \langle \hat{N}_{ij}, N_{ij} \rangle), \quad (13)$$

where $\langle \cdot, \cdot \rangle$ denotes the dot product, $\hat{N}_{ij}$ represents the estimated surface normal at pixel position (i, j) , and N_{ij} is the ground truth surface normal corresponding to the surface. By minimizing the loss, we can effectively enhance the model's accuracy in reconstructing surface normal.

4. Experimental results and discussion

Implementation details. The network model was constructed and trained using the PyTorch framework. The training process was carried out on two NVIDIA RTX 3090 GPUs (24 GB) with a batch size of 16, over a total of 1000 epochs. The Adam optimizer was selected, and the initial learning rate was set to $1e-4$ to balance training speed and convergence effectiveness. To enhance the model's generalization ability and robustness, a data augmentation strategy was employed. During each training iteration, the input images were randomly cropped into 512×512 patches, thereby increasing the diversity of the training data and reducing the risk of overfitting.

Evaluation metrics. To comprehensively assess the performance of the proposed method in surface normal estimation, this study references prior related works [3,7,16] and employs six metrics to quantitatively analyze the accuracy of the predicted normals. These metrics include three angular error measures: mean angular error (Mean_ae = $\frac{1}{N} \sum_{i=1}^N \theta_i$), median angular error (Median_ae = median($\theta_1, \theta_2, \dots, \theta_N$)), and root mean square angular error (RMSE_ae = $\sqrt{\frac{1}{N} \sum_{i=1}^N \theta_i^2}$), which represent the average, median, and root mean square of all

valid pixel angular errors, respectively. Lower values for these metrics indicate better performance. Where N is the total number of effective pixels, $\theta_i = \frac{180}{\pi} \arccos \left(\frac{\mathbf{n}_{\text{pred},i} \cdot \mathbf{n}_{\text{gt},i}}{\|\mathbf{n}_{\text{pred},i}\| \|\mathbf{n}_{\text{gt},i}\|} \right)$, which represents the angular error of pixel i , where $\mathbf{n}_{\text{pred}}$ and $\mathbf{n}_{\text{gt}}$ represent predicted surface normals and ground truth respectively. Additionally, three accuracy ratios are utilized, representing the proportions of valid pixels with angular errors below 11.25° ($\text{ae_11} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(\theta_i < 11.25^\circ)$), 22.5° ($\text{ae_22} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(\theta_i < 22.5^\circ)$), and 30.0° ($\text{ae_30} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(\theta_i < 30^\circ)$), where $\mathbb{I}(\cdot)$ is the indicator function (returning 1 if the condition is satisfied and 0 otherwise). Higher values of these metrics indicate better reconstruction accuracy.

The first three angular error metrics are primarily used to quantify the angular deviation between the predicted normal and the ground truth normal, providing a detailed error analysis. The subsequent three accuracy ratios measure the percentage of pixels for which the predicted normal fall within the specified angular thresholds (11.25° , 22.5° , and 30.0°), thereby reflecting the model's performance under varying precision requirements. Through this comprehensive evaluation framework, the present study is able to thoroughly demonstrate the accuracy and reliability of the proposed method's normal estimation across different scenarios and conditions.

4.1. Dataset

Two representative datasets, namely DeepSfP [16] and SPW [3], were selected from publicly available real-world SfP datasets for experimental validation in this study.

DeepSfP: DeepSfP is the first publicly available object-level SfP dataset, where each image contains only a single object while providing high-quality surface normals. We adopt the training/testing split provided in the original paper [16]. **SPW:** SPW is the first publicly available real-world scene-level SfP dataset, comprising 522 image sets captured across 110 distinct real-world scenes. Each scene contains multiple target objects composed of different materials with varying reflective properties, featuring multiple highly reflective areas within a single scene to enable investigation of the relationship between SfP and material characteristics. In the original paper [3], 403 sets were allocated for training and 119 for testing. However, the publicly released version of the SPW dataset only provides 110 test sets. Consequently, we ultimately adopted 403 sets for training and 110 sets for testing.

4.2. Comparison to SfP baselines

Comparative experiments were conducted on the DeepSfP and SPW datasets using three physics-based methods (Miyazaki et al. [26], Mahmoud et al. [8], Smith et al. [9]) and four deep learning-based methods (Ba et al. [16], Kondo et al. [17], Lei et al. [3], Liu et al. [7]). The original code and default parameters released in [9] were utilized to obtain the surface normal estimation results from the three physics-based methods. Since Kondo et al. [17] and Liu et al. [7] did not publish their source code, the evaluation metrics employed by Liu et al. [7] on the DeepSfP and SPW datasets were used. In the Ba et al. [16] model, due to the lack of evaluation metrics in the article. So we retrained on two datasets and obtained corresponding evaluation metrics. Regarding the method proposed by Lei et al. [3], due to the mismatch in the number of training and testing sets and the number of available images in the released SPW dataset, the published source code and parameters provided in the paper were used to retrain the model on both datasets.

Qualitative comparisons were conducted on the DeepSfP dataset, and the visualization results for four objects with surface textures ranging from simple to complex are presented in Fig. 6. Among them, the vase features a relatively simple surface texture and was used to demonstrate the differences in the ability of each method to reconstruct smooth surface normal. It can be clearly observed from Fig. 6 that significant angular errors exist in the reconstruction results of the three physics-based methods. In contrast, because polarization and ambiguous normal information

were combined in the proposed method, the reconstruction results surpassed those of the other four deep learning methods in both smoothness and accuracy, particularly in the magnified regions. The remaining three objects possess complex texture details, making it difficult for purely physics-based methods to effectively estimate surface normal. The methods by Ba et al. [16] and Kondo et al. [17] were only able to reproduce texture details in a somewhat ambiguous manner. Although the methods by Lei et al. [3] and Liu et al. [7] were capable of reconstructing clearer details, some false textures were also generated. In comparison, the proposed method ensures the smoothness of the reconstructed surfaces while maintaining accurate restoration of details. Table 1 provides the quantitative results of all methods on the DeepSfP dataset. It is clearly evident that deep learning-based methods hold a greater advantage over traditional physics-based methods. Among all the compared methods, the proposed method achieved the best performance in terms of angular error and accuracy.

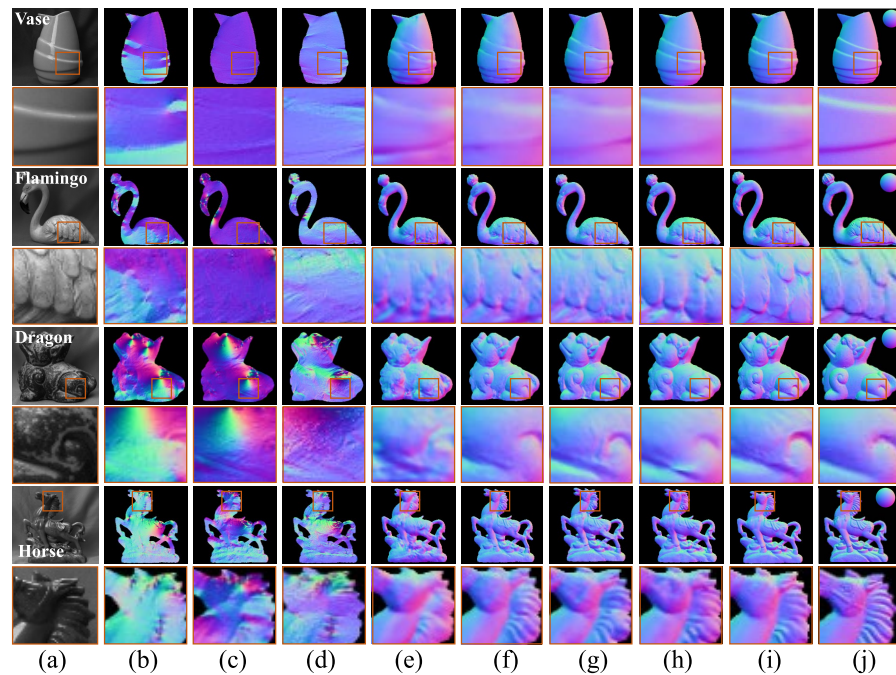


Fig. 6. Visualization results comparing on the DeepSfP dataset. (a) Four objects with shapes and textures ranging from simple to complex are represented. (b) Miyazaki et al. [26] (c) Mahmoud et al. [8] (d) Smith et al. [9] (e) Ba et al. [16], (f) Kondo et al. [17] (g) Lei et al. [3] (h) Liu et al. [7] (i) Our proposed method. (j) The ground truth. The large boxes correspond to magnified views of the small boxes. The spheres inserted into the ground truth are color-visualized to show the direction of the surface normal.

Additionally, comparative experiments were conducted on the SPW dataset with real-world scenes and selected five representative scenarios for visualization, as shown in Fig. 7. Among the five scenarios, three are indoor (Scene1-Scene3) and two are outdoor (Scene4 and Scene5). Each scene contains objects with at least three different material properties (i.e., different refractive indices n and reflectance η). For example, Scene3 includes walls, floors, and trash bins, each with distinct materials. The visualization results are shown in Fig. 7. For the SPW dataset, which features real-world environments with multiple discontinuous surfaces and higher scene complexity, physics-based methods struggle to achieve effective reconstruction results. Among the deep learning-based SfP methods: Ba et al.'s approach has difficulty handling discontinuous

Table 1. Quantitative evaluation on the DeepSfP dataset. The best and second-best results are shown in red and blue, respectively.

Method	Description	Angular Error ↓			Accuracy ↑		
		Mean	Median	RMSE	11.25°	22.5°	30.0°
Miyazaki et al.	Physics-based SfP	44.83	40.66	49.65	10.23	15.81	23.17
Mahmoud et al.	Physics-based SfP	50.93	50.03	52.09	8.99	13.67	21.31
Smith et al.	Physics-based SfP	46.48	47.22	48.57	16.72	29.91	39.46
Ba et al.	Learning-based SfP	18.15	14.35	22.81	37.86	72.21	83.54
Kondo et al.	Learning-based SfP	14.74	12.25	18.10	51.33	79.97	88.97
Lei et al.	Learning-based SfP	14.44	10.69	18.91	52.64	80.42	89.54
Rui Liu et al.	Learning-based SfP	13.52	10.68	16.97	56.65	81.33	90.06
Ours	Learning-based SfP	12.61	9.19	16.79	59.50	85.30	92.18

surfaces, resulting in holes and loss of texture details; Kondo et al.'s method can distinguish surface normals but produces blurry texture reconstructions; Lei et al.'s approach is prone to normal errors in specular reflection regions (e.g., the wall in Scene3); and Liu et al.'s method underperforms in areas with abrupt changes in continuous surfaces (e.g., the cabinet door in Scene2). In contrast, our proposed method maintains high-quality reconstruction across real-world scenes with diverse material properties and reflection types (diffuse and specular). Table 2 presents the quantitative results of all methods on the SPW dataset. It is evident that deep learning-based methods significantly outperform traditional physics-based approaches. Among all compared methods, our approach achieves the best performance in terms of angular error and accuracy.

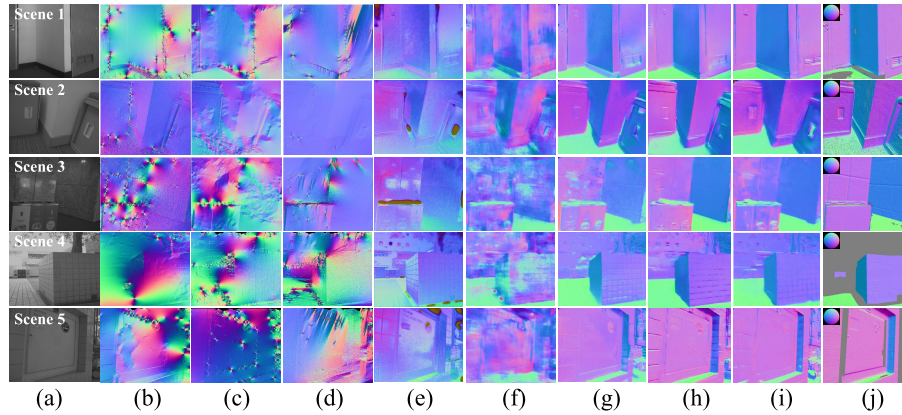


Fig. 7. Visualization results comparing on the SPW dataset. (a) Five different scenes are represented. (b) Miyazaki et al. [26] (c) Mahmoud et al. [8] (d) Smith et al. [9] (e) Ba et al. [16], (f) Kondo et al. [17] (g) Lei et al. [3] (h) Liu et al. [7] (i) Our proposed method. (j) The ground truth. The spheres inserted into the ground truth are color-visualized to show the direction of the surface normal.

Finally, we evaluated our model's parameter count and inference speed on an NVIDIA RTX 3090 GPU (24GB). As shown in Table 3, compared to the other two deep learning-based SfP methods, our proposed dual-branch deep fusion network exhibits certain disadvantages in terms of parameter count and inference speed. This performance shortcoming is an inevitable trade-off brought about by the design of dual-branch network architecture, but offers unique advantages in feature fusion effectiveness.

Table 2. Quantitative evaluation on the SPW dataset. The best and second-best results are shown in red and blue, respectively.

Method	Description	Angular Error ↓			Accuracy ↑		
		Mean	Median	RMSE	11.25°	22.5°	30.0°
Miyazaki et al.	Physics-based SfP	54.01	51.60	59.69	3.97	13.87	22.37
Mahmoud et al.	Physics-based SfP	53.13	51.93	57.02	2.98	12.37	21.31
Smith et al.	Physics-based SfP	49.35	47.26	51.87	11.33	24.86	34.54
Ba et al.	Learning-based SfP	39.98	34.34	40.33	8.14	28.50	43.82
Kondo et al.	Learning-based SfP	29.49	25.91	34.63	16.95	45.51	60.82
Lei et al.	Learning-based SfP	18.27	14.49	23.20	41.91	74.62	84.44
Rui Liu et al.	Learning-based SfP	17.85	14.35	22.75	44.45	75.37	84.62
Ours	Learning-based SfP	16.56	12.62	21.67	50.60	79.38	87.11

Table 3. The model's parameter size and inference speed.

Method	Description	Model Parameter	Inference Speed
Ba et al.	Learning-based SfP	10.93M	1.491s
Lei et al.	Learning-based SfP	42.48M	1.658s
Ours	Learning-based SfP	93.17M	2.463s

4.3. Importance of ambiguous surface normals

To validate the enhancement effect of the ambiguous surface normals, calculated using the pathological physical model, on surface normal reconstruction, we designed real-world comparative experiments on the SPW dataset. Three representative samples with complex scenes were selected. These three samples include walls with multiple specular reflection regions and various material properties, such as wall surfaces, switches, iron barrels, and floors. Figure 8 shows the reconstruction results and error maps of the three target scenes we designed. The model by Ba et al. [16] fails to effectively reconstruct scene-level targets, resulting in severe errors in the reconstructed surface normal, with large discrepancies. On the other hand, Lei et al.'s [3] method optimizes scene targets by incorporating efficient phase angle encoding, multi-head attention, and viewpoint encoding information. However, due to the network input being solely the polarization prior information, considerable reconstruction errors still remain, especially in regions with dim reflected light (such as the blackboard in the first two scenes) and high-reflection areas (such as the walls in all three scenes), leading to significant errors.

To highlight the importance of the physical model prior features for surface normal reconstruction, we removed the ambiguous surface normals and specular confidence features and input only the polarization prior, using a single-branch CGA-based network structure, as shown in the fourth column of Fig. 8. The surface normal reconstructed by the single-branch CGA network with polarization prior still exhibit considerable errors. To verify the significance of the ambiguous surface normals calculated from the pathological physical model, we input both the polarization prior and the ambiguous surface normals into the network via the CGA-based hybrid fusion module, as shown in the fifth column of Fig. 8. The inclusion of the ambiguous surface normals significantly reduced the overall error, demonstrating the importance of the ambiguous surface normals as network input. However, due to the absence of specular confidence information in the network, the specular reflection regions on the walls in the three scenes still show considerable errors in the fifth column. Therefore, we further added the specular confidence H to verify its importance. The sixth column in Fig. 8 shows the results of our proposed method. After adding the specular reflection regions, the errors in the specular areas were eliminated, and the

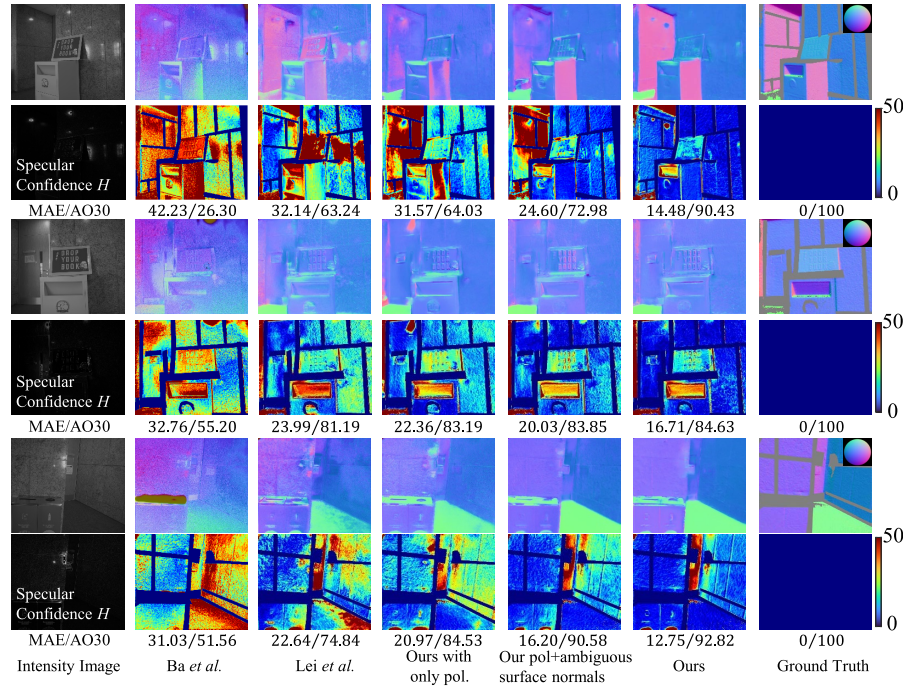


Fig. 8. The error maps of the importance of ambiguous surface normals are evaluated using both the mean angular error (MAE) and the accuracy over 30° (AO30). The error maps are rendered with a color scheme converted to JET space. The spheres inserted in the ground truth are color-coded to visualize the orientation of surface normals.

reconstructed surface normal in those regions were accurate and smooth, further confirming the effectiveness of the specular confidence map H .

The reconstruction errors were evaluated by introducing the mean angular error and accuracy over 30° . As shown in Fig. 8, compared to the single-branch CGA network with only polarization priors, our method achieves metric improvements of 17.09/26.4%, 5.65/1.44%, and 8.22/8.29% in the three scenes. When compared to the dual-branch CGA hybrid fusion network that includes both polarization and ambiguous surface normals, our proposed method shows improvements of 10.12/17.45%, 3.32/0.78%, and 3.45/2.24%, respectively. Rows 5, 6, and 7 of Table 4 list the six evaluation metrics for all imaging scenes in the SPW dataset, demonstrating that the incorporation of ambiguous surface normals information significantly improves surface normal reconstruction performance.

4.4. Ablation studies

The ablation experiments of the proposed method were conducted on the SPW dataset. All methods utilized the same random seed and an initial learning rate of $1e-4$. Firstly, the importance of each component in the polarization prior was validated through four ablation experiments: (i) removal of I_{un} ; (ii) removal of ϕ_e ; (iii) removal of ρ ; and (iv) removal of V . The quantitative comparison results are presented in the first to fourth rows of Table 4. It can be observed that the exclusion of any component of the polarization prior leads to a decline in the fusion performance, which underscores the rationality of constructing the polarization prior. Secondly, three ablation experiments were conducted on the ambiguous surface normals and specular confidence prior information computed from the ill-conditioned physical model: (i) removal of both ambiguous surface normals (ASN) and specular confidence information

Table 4. Results of ablation experiments on the SPW Dataset. The optimal outcomes are highlighted in red.

Method	Angular Error ↓			Accuracy ↑		
	Mean	Median	RMSE	11.25°	22.5°	30.0°
w/o I_{in}	21.74	19.00	26.41	27.47	59.70	76.36
w/o ϕ_e	18.15	14.69	22.44	36.77	71.72	83.53
w/o ρ	17.42	13.97	21.58	38.03	74.26	85.18
w/o V	17.23	13.80	21.45	39.71	73.95	84.76
w/o ASN and H	20.57	17.68	24.51	26.69	64.19	79.26
w/o ASN	19.54	15.61	24.38	34.02	68.69	81.25
w/o H	19.27	15.38	24.43	34.69	69.49	83.11
w/o Transformer Encoder	18.82	14.93	24.01	37.73	71.96	77.64
w/o CGA	17.44	13.53	22.51	41.01	74.56	80.10
w/o CGA-based hybrid fusion scheme	21.98	11.59	26.67	25.40	62.69	77.51
Ours	16.56	12.62	21.67	50.60	79.38	87.11

(H); (ii) removal of ASN; and (iii) removal of H . The quantitative comparison results are shown in the fifth to seventh rows of Table 4. It is evident that the removal of ambiguous surface normals and specular confidence information results in decreased reconstruction accuracy, highlighting the significance of the proposed ambiguous surface normals and specular confidence information in assisting the polarization prior to reconstruct high-quality surface normal. Finally, to validate the effectiveness of our proposed CGA-Transformer network, we conducted three ablation experiments: (i) removing the Transformer Encoder; (ii) removing the CGA module; and (iii) removing the CGA-based hybrid fusion module. The quantitative results are shown in rows 8-10 of Table 4. The quantitative results indicate that removing any module would degrade the reconstruction accuracy of surface normals, demonstrating the effectiveness of our network.

5. Conclusion

A novel deep learning-based SfP method is proposed in this paper, which employs a CGA-based hybrid fusion scheme to effectively integrate prior knowledge such as polarization information, ambiguous surface normals, and specular confidence. The ambiguous surface normals information provides additional support to the polarization cues, while the specular confidence is specifically utilized to address errors in surface normal reconstruction under complex lighting and specular reflection environments. Furthermore, a multi-head attention mechanism is introduced to capture the global feature information of the images, thereby achieving high-precision and robust reconstruction of surface normal. Compared to existing state-of-the-art deep learning-based SfP methods, our approach demonstrates superior performance in both qualitative and quantitative evaluations, achieving high-quality 3D reconstructions on both object-level and scene-level datasets.

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Data availability. Data will be made available on request.

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